

llmXive Automated Review of VLA-Corrector: Lightweight Detect-and-Correct Inference for Adaptive Action Horizon

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a coherent set of claims regarding the VLA-Corrector framework, with most quantitative assertions supported by the provided tables and figures. However, there are specific instances where the text’s summary of results does not align with the detailed data tables, leading to potential confusion or overstatement of efficiency gains.

The most significant discrepancy is in the Introduction (Section 1), where the authors claim a “+24.6% success-per-call efficiency gain” for the $\pi_{0.5}$ model at horizon 50. The text cites a success rate increase from 48.7% to 58.7% and a call reduction from 5.15 to 4.98. A direct calculation of the efficiency gain—defined as (Success/Calls) for the new method divided by the baseline—yields approximately 20.0%, not 24.6%. Furthermore, the success rate of 58.7% cited in the text corresponds to the “Horizon 50” row in Table 3, whereas the “Avg” column in Table 1 (which the text seems to reference for the baseline 48.7%) shows a different result (64.35% for the full method). This conflation of specific horizon results with average results creates a factual inconsistency in the narrative.

Additionally, the claim that the method “surpasses” the fully fine-tuned baseline on LIBERO (97.8% vs 96.9%) is numerically true based on Table 2, but the margin is narrow (0.85%). While not an error, the phrasing “surpassing” without qualification might overstate the robustness of this specific finding to a reader expecting a larger margin of improvement.

Finally, the claim regarding the 83.7% truncation rate in “critical phases” relies on a manual, subjective classification of trajectory phases. While the number is internally consistent with the paper’s own analysis, the lack of an objective, reproducible definition for “critical” makes this specific statistic difficult to verify or replicate, though it does not constitute a fabrication.

Overall, the paper’s core scientific claims are supported by the data, but the narrative summary in the Introduction requires correction to accurately reflect the numbers in the results tables.

Action items.

- **[writing]** The paper presents a coherent set of claims regarding the VLA-Corrector framework, with most quantitative assertions supported by the provided tables and figures. However, there are specific instances where the text’s summary of results does not align with the detailed data tables, leading to potential confusion or overstatement of efficiency gains. The most significant discrepancy is in the Introduction (Section 1), where the authors claim a “+24.6% success-per-call efficiency gain” for the $\pi_{0.5}$

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure demonstrates a moving-object grasp but contains a discrepancy between the caption (mentioning a white bowl) and the visual evidence (showing a clear cup). Additionally, the top row is cluttered with large text and graphics that obscure the scene.

Figure 2

The figure demonstrates a successful manipulation task, but the visual evidence contradicts the caption’s claim regarding the target container (blue vs. clear bowl).

Figure 3

The figure demonstrates the moving-insertion task but suffers from poor presentation, specifically the informal text overlay and lack of clear temporal markers to track the robot’s reaction to the disturbance.

Figure 4

Figure 4 effectively illustrates the contrast between open-loop and closed-loop execution using a clear side-by-side layout. The visual elements, including the timeline, action chunks, and outcome indicators, are well-labeled and align perfectly with the provided caption.

Figure 5

Figure 5 is a clear and well-structured schematic that effectively visualizes the VLA-Corrector pipeline. The four distinct blocks (A-D) logically follow the caption’s description, and the comprehensive legend on the right clearly defines the color-coded tokens and latent variables used throughout the diagram.

Figure 6

The figure is a clear trade-off plot, but the caption describes a two-panel layout (‘Left’ and ‘Right’) while only a single plot is rendered, omitting the data required to support the efficiency gain claim. Additionally, the model name in the caption is malformed.

Figure 7

The figure effectively visualizes the distribution of the inconsistency score E_t for success vs. failure, but the right panel’s caption is misleading as it describes ‘interrupt frequency’ while the plot actually shows the mean score distribution.

Figure 8

The figure fails to support its caption’s claims; the left panel shows static task phases rather than truncation frequency data, and the right panel containing the cited statistics is missing entirely.

Figure 9

The figure effectively demonstrates the qualitative difference between the baseline and the proposed method using color-coded bounding boxes. However, it lacks an internal legend to define the ‘INTERRUPT’, ‘FAILURE’, and ‘SUCCESS’ labels, and does not explicitly label the rows corresponding to the methods described in the caption.

Figure 10

The figure attempts to illustrate a real-world disturbance recovery but lacks visual annotations to clarify the sequence of events, and the timing of the bowl shift appears inconsistent with the caption’s description of recovery during execution.

Action items.

- **[science]** Figure 1: The caption states the robot places the cube into the ‘white bowl’, but the images show the robot placing the cube into a clear plastic cup, while a white bowl is visible in the background but not used.
- **[writing]** Figure 1: The top row contains a large text overlay (‘Change the cube’s position’) and a graphical arrow that obscure the view of the robot and the scene, reducing the clarity of the visual demonstration.
- **[science]** Figure 2: The caption states the robot places the cube into the blue bowl, but the visual sequence shows the robot placing the cube into the clear (transparent) bowl in the final frames.
- **[writing]** Figure 3: The text overlay ‘drawer table moved’ is unprofessional and lacks a clear legend or arrow definition explaining the specific perturbation event compared to the other demo figures.
- **[science]** Figure 3: The sequence of images is not clearly labeled with timestamps or step numbers, making it difficult to verify the chronological order of the ‘moving-insertion’ task and the robot’s reaction to the drawer movement.
- **[science]** Figure 6: The caption claims the figure contains a ‘Right’ panel showing ‘success-per-call efficiency’, but the rendered image displays only a single plot. The missing panel

prevents verification of the ‘consistent efficiency gains’ claim.

- **[writing]** Figure 6: The caption contains a formatting error in the model name, written as ‘ $\pi_{0.5}$ ’ instead of the likely intended ‘ $\pi_{0.5}$ ’ (matching the figure legend).
- **[science]** Figure 7: The right panel’s caption claims to show ‘interrupt frequency,’ but the y-axis is labeled ‘Episode mean E_t ’ and the plot displays violin distributions of the score itself, not a frequency count of interrupts.
- **[science]** Figure 8: The caption claims the left panel shows ‘representative trajectories’ where truncation is triggered more frequently, but the image displays static snapshots of task phases (Grasp, Align, Place, etc.) without any data visualization (e.g., plots, heatmaps, or frequency markers) to demonstrate truncation frequency.
- **[science]** Figure 8: The caption states the right panel shows ‘83.7% of truncations occur in critical phases’, but the rendered image contains no right panel, chart, or statistical graphic to support this quantitative claim.
- **[writing]** Figure 9: The image contains red and green bounding boxes with text labels (‘INTERRUPT’, ‘FAILURE’, ‘SUCCESS’) but lacks a formal legend or key to define these visual indicators.
- **[writing]** Figure 9: The caption describes the top row as a ‘monitored baseline’ and the bottom as ‘VLA-Corrector’, but the image itself does not label the rows, relying entirely on the caption for this distinction.
- **[science]** Figure 10: The caption claims a human shifts the blue bowl during execution, but the image shows the bowl moving from the right to the center while the robot arm is already positioned over the center, suggesting the disturbance occurred before the robot reached the target rather than during the grasp attempt.
- **[writing]** Figure 10: The image contains no labels, arrows, or annotations to indicate the timing of the disturbance or the robot’s reaction, making it difficult to verify the ‘recovery’ claim visually.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-written and accessible to a competent reader in robotics or machine learning, with most acronyms (VLA, LVM, OGG) defined at first use. However, there are specific instances of notation overload and undefined symbols that could stall a reader from an adjacent field (e.g., a control theorist or a researcher in standard RL who is less familiar with flow-matching specifics).

First, in Section 3.4, the symbol τ is introduced as the “denoising step” in the context of flow matching. While standard for diffusion/flow model practitioners, the paper does not explicitly state the domain of τ (e.g., continuous $[0, 1]$ or discrete steps) or the specific flow-matching formulation being used. A reader from a different subfield might struggle to map this to the velocity field update in Eq. 8. A brief parenthetical definition would resolve this.

Second, there is a potential notation collision in Section 3.4. The symbol ΔZ_t^* is used to denote the “accumulated deviation” ($Z_t^{\text{real}} - Z_{t-k}^{\text{real}}$). However, in Section 3.1 (Eq. 1), the notation ΔZ_{t+k}^* is explicitly defined as the *target* residual from demonstrations (ground truth). Using the same starred notation for an *observed* drift in the OGG section creates ambiguity: is this the ground truth target or the actual observed difference? The text implies the latter, but the notation suggests the former. Renaming the observed drift term (e.g., to ΔZ_{obs} or ΔZ_{dev}) would eliminate this confusion.

Finally, the term “flow matching” and “velocity field” are used in Section 3.1 and 3.4 without a brief operational definition. While these are standard in the generative AI community, a reader from a traditional control or standard RL background might not immediately grasp that this refers to a specific generative modeling technique where a network predicts a velocity vector field. A single clarifying clause would ensure the method is self-contained for the broader robotics community.

These are minor fixes that significantly improve the self-containment of the paper for the target “adjacent-field PhD” audience.

Action items.

- **[writing]** Section 3.4 (OGG Correction) introduces the symbol $\mathbf{au}$ as ‘denoising step’ without defining the flow-matching context or the range of $\mathbf{au}$. An adjacent-field reader may not know if $\mathbf{au} \in [0, 1]$ or discrete steps. Add a brief clause: ‘where $\mathbf{au} \in [0, 1]$ is the flow-matching time step’.
- **[writing]** Section 3.4 defines $\Delta Z_{\text{dev}} = \Delta Z_t^* = Z_t^{\text{real}} - Z_{t-k}^{\text{real}}$ but uses the notation ΔZ_t^* (with a star) which was previously defined in Eq. 1 as the target residual from demonstrations. This overloads the symbol: in Eq. 1 it is a ground-truth target, here it is an observed drift. Clarify the distinction or use a distinct symbol (e.g., ΔZ_{obs}) for the observed drift to avoid confusion.
- **[writing]** Section 3.3 introduces the patience parameter p in Eq. 5 without defining its role or typical value range before the equation. While defined later in the appendix, the main text should briefly state: ‘where p is a patience threshold (e.g., 5 steps) to filter transient spikes’.
- **[writing]** Section 3.1 uses the term ‘flow matching’ and ‘velocity field’ without a one-sentence gloss for readers from non-generative-model backgrounds (e.g., standard RL or control). While standard in generative AI, an adjacent control theorist might need a brief operational definition: ‘a generative process that learns a velocity field to transport noise to data’.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is generally sound, but two specific logical inconsistencies regarding the definition of the baseline and the causal mechanism for efficiency gains require clarification.

First, in Section 5.2 (“Mechanism Analysis”), the description of the qualitative comparison in Figure 4 contains a definition error. The text states: “In the top row, the baseline rollout is only monitored by LVM for error alignment without truncating the remaining actions.” This is logically inconsistent with the paper’s own definition of the “Baseline” (e.g., in the Abstract and Section 1), which is a standard action-chunked policy *without* the Latent-space Vision Monitor (LVM). The baseline cannot be “monitored by LVM” because LVM is the novel component of the proposed method. The authors likely intended to compare the “Baseline” (no LVM, no truncation) against “VLA-Corrector” (LVM + truncation), or they are comparing against a specific ablation variant (e.g., “LVM only, no truncation”) but have mislabeled it as “the baseline.” This conflation of the standard baseline with a partial-ablation variant undermines the clarity of the causal claim that “truncation” is the key differentiator in that specific figure.

Second, in Section 5.1 (“Main Results”), the text claims that VLA-Corrector “reduces policy-call frequency” for SmolVLA at horizon 10, citing a drop from 19.27 to 15.64 calls per episode. This conclusion does not logically follow from the premises of the method as described. The VLA-Corrector mechanism is defined as an *interrupt-triggered* system: it monitors execution and, upon detecting drift, *truncates* the current chunk and *invokes* a new policy call (replanning). Intuitively, adding an interrupt handler that triggers *additional* replans should increase, not decrease, the total number of policy calls compared to a fixed-horizon baseline that simply executes the chunk. The paper does not provide a premise explaining *why* the baseline would require *more* calls (e.g., does the baseline fail and restart the entire episode, whereas VLA-Corrector recovers mid-episode?). Without this explicit premise, the conclusion that the method “reduces policy calls” is a non-sequitur based on the provided text. The authors must either clarify the counting methodology (e.g., “calls per successful episode” vs. “calls per attempt”) or rephrase the claim to reflect that the method improves “success-per-call” efficiency (which is supported by the data) rather than reducing the raw call count.

Action items.

- **[writing]** Section 5.1 claims VLA-Corrector ‘reduces policy calls’ (19.27 to 15.64) for SmolVLA, yet the method adds interrupt-triggered replans. The text lacks a premise explaining how adding replans reduces total calls (e.g., baseline episode restarts vs. mid-episode recovery). Clarify the counting logic or rephrase to ‘improves success-per-call’.
- **[writing]** Section 5.2 states the ‘baseline rollout is only monitored by LVM,’ but the baseline is defined elsewhere as lacking LVM. This contradicts the method definition. Clarify if the

comparison is against an ‘LVM-only’ ablation or correct the description to ‘Baseline (no LVM)’.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper generally maintains a strong alignment between its experimental evidence and its claims, particularly in the detailed ablation studies and the clear distinction between simulation and real-world results. However, there are three instances where the rhetoric slightly exceeds the specific scope of the demonstrated evidence, primarily regarding the universality of the “contact-rich” claim and the uniformity of performance gains.

First, the Abstract asserts that the method improves robustness in “contact-rich robotic manipulation tasks” as a general category. While the real-world experiments (Appendix D) do involve contact-rich tasks (insertion, alignment), they are conducted exclusively on a single robot platform (AgileX PiPER) with a specific set of nine tasks. The paper does not demonstrate transfer to other robot morphologies (e.g., 7-DoF arms, mobile manipulators) or different contact dynamics (e.g., fluid handling, deformable objects). The claim should be scoped to the tested domain or accompanied by a limitation statement acknowledging the single-platform evaluation.

Second, the Introduction states that “Gains are consistent across all three backbones.” While the direction of improvement is consistent (positive), the magnitude varies significantly: $\pi_{0.5}$ sees a +15.65% gain, while X-VLA sees only +4.05%. The word “consistent” can misleadingly imply uniform efficacy. A more precise phrasing would be “Gains are observed across all three backbones,” allowing for the variance shown in Table 1.

Finally, the Conclusion presents the adaptive logic as a definitive solution (“long-horizon execution is preserved when reliable...”). However, the paper’s own failure analysis (Appendix D.4) explicitly identifies scenarios where the method fails: when disturbances exceed the robot’s reachable workspace or when the frozen backbone policy lacks the capacity to generate a valid recovery trajectory. The conclusion should acknowledge these hard limits, noting that the method improves robustness *within the representational and physical bounds of the underlying policy*, rather than implying it solves the open-loop blind spot universally.

These are minor rhetorical adjustments that would bring the paper’s framing into perfect alignment with its empirical boundaries.

Action items.

- **[writing]** The paper generally maintains a strong alignment between its experimental evidence and its claims, particularly in the detailed ablation studies and the clear distinction between simulation and real-world results. However, there are three instances where the rhetoric slightly exceeds the specific scope of the demonstrated evidence, primarily regarding the universality of the “contact-rich” claim and the uniformity of performance gains. First,

the Abstract asserts that the method improves robustn

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a safety-enhancing framework (VLA-Corrector) designed to improve the robustness of Vision-Language-Action (VLA) models in physical robotics by detecting and correcting execution drift. The methodology involves a Latent-space Vision Monitor (LVM) and Online Gradient Guidance (OGG) to truncate stale action chunks and replan when deviations occur.

From a safety and ethics perspective, the work is low-risk and arguably beneficial. The primary contribution is mitigating the “open-loop blind spot” where robots might continue executing erroneous actions, potentially leading to collisions or task failure. By introducing a mechanism to detect and correct these errors, the paper directly addresses a safety concern inherent in current action-chunking paradigms.

The research relies on standard simulation benchmarks (MetaWorld, LIBERO) and a controlled real-world setup using an AgileX PiPER arm with teleoperated demonstrations. The data sources are public benchmarks or self-collected demonstrations for fine-tuning, with no indication of using private, sensitive, or non-consented human data. The real-world experiments involve human disturbances applied in a controlled manner to test recovery, which is a standard safety validation protocol and does not pose a risk of harm to subjects or the environment.

There are no dual-use concerns; the method is specific to robotic control stability and does not lower the barrier for generating harmful content, cyberattacks, or biological hazards. The paper does not release any PII, and the code is hosted on a public repository as is standard practice. No conflicts of interest are apparent beyond standard academic affiliations.

As this is a third-party preprint, the absence of a formal IRB statement is noted, but given the use of public benchmarks and standard teleoperation protocols for data collection (which typically fall under exempt categories or standard lab safety protocols rather than human-subjects research requiring full IRB review), this does not constitute a fatal flaw or a missing disclosure of significant risk. The paper successfully identifies a safety risk (error accumulation) and proposes a mitigation without introducing new, unaddressed risks.

Verdict: Accept. No action items required.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling framework for adaptive action horizons, but the experimental design in the main results tables lacks the necessary statistical rigor to fully support the magnitude of the claimed improvements.

1. Missing Variance and Seed Counts in Main Benchmarks The central claims regarding performance gains on MetaWorld (Table 1) and LIBERO (Table 2) rely on single-point success rates (e.g., 64.35% vs 48.70%). The paper mentions in the appendix that multi-seed data-scaling analyses used seeds {42, 123, 999}, but it does not explicitly state that the *main* benchmark results in Tables 1 and 2 were averaged over these seeds, nor does it report the standard deviation. In VLA research, success rates can vary significantly (often $\pm 3\text{-}5\%$) across seeds due to initialization and stochasticity. A 15-point gain is likely real, but without reported variance (mean $\pm$ SD) and a clear statement of the number of seeds (n), a skeptical reader cannot rule out that the reported numbers are the result of a “lucky seed” or an outlier run. The design currently fails to distinguish a robust effect from sampling noise.

2. Aggregated Variance in Real-World Results In the real-world evaluation (Table 3), the authors report “Mean $\pm$ SD across tasks” (e.g., 73.3 ± 6.5). This is a methodological flaw in the evidence design. Aggregating variance across three distinct task groups (Pick-and-place, Alignment, Disturbance) with different difficulty levels obscures the true reliability of the method on any single task. A method could have high variance on the “Disturbance” tasks (where it is most needed) but low variance on “Pick-and-place,” resulting in a misleadingly low overall SD. To support the claim of robustness, the paper must report the success rate and standard deviation *across the 20 trials* for each individual task, not just the average across tasks.

3. Baseline Tuning Asymmetry The ablation study in Table 4 isolates the components (Truncation vs. OGG), but the comparison against the “Baseline” in the main tables (Table 1) does not explicitly confirm that the baseline policies were subjected to the same hyperparameter search (e.g., learning rate, action horizon tuning) as the VLA-Corrector variants. If the baseline was a “vanilla” run while the proposed method benefited from implicit tuning during the development of the corrector, the reported gains may be confounded by the tuning effort rather than the mechanism itself. The evidence design requires a “fair baseline” control where the baseline is re-tuned with the same budget to isolate the contribution of the detect-and-correct mechanism.

Addressing these points by reporting seed counts, per-task trial variance, and ensuring fair baseline tuning will significantly strengthen the evidentiary support for the paper’s central claims.

Action items.

- **[writing]** The paper presents a compelling framework for adaptive action horizons, but the experimental design in the main results tables lacks the necessary statistical rigor to fully support the magnitude of the claimed improvements. 1. Missing Variance and Seed Counts in Main Benchmarks The central claims regarding performance gains on MetaWorld (Table 1) and LIBERO (Table 2) rely on single-point success rates (e.g., 64.35% vs 48.70%). The

paper mentions in the appendix that multi-seed data-scaling anal

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical reporting in this paper is generally descriptive but lacks the necessary uncertainty quantification required to validate the magnitude of the reported improvements.

First, the main simulation results in **Table 1** (MetaWorld) and **Table 2** (LIBERO) present success rates as precise point estimates (e.g., “64.35%”) without any accompanying measure of variance (standard deviation, standard error) or indication of the number of random seeds used. In reinforcement learning and robotics, performance is inherently stochastic due to initialization, environment dynamics, and policy sampling. Reporting a single number without a spread (e.g., mean $\pm$ SD over 5 seeds) creates an illusion of false precision and prevents readers from assessing the stability of the method. The authors should aggregate results over multiple seeds and report the mean and standard deviation for all primary comparisons.

Second, the real-world results in **Table 3** (Appendix) and the main text calculate the “Group Avg” and “Overall Avg” by taking the mean of the task-level success rates and then reporting the standard deviation of those *task means* (e.g., “73.3 $\pm$ 6.5”). This is a statistical misinterpretation for the purpose of claiming method performance. The standard deviation of task means reflects the *difficulty variance* of the benchmark tasks, not the *reliability* of the VLA-Corrector method. To correctly report the precision of the average success rate, the authors should either report the standard deviation of the 20 trials within each task (aggregated appropriately) or, more standardly, report the standard error of the mean (SEM) across the 9 tasks to indicate how precisely the 73.3% figure estimates the true population performance.

Finally, the claim in **Section 5.2** that OGG yields an “average gain of 0.23” in recovery rates (referencing Figure 5) is presented without statistical backing. While the visual trend is clear, a quantitative claim of “0.23” implies a specific effect size that should be accompanied by a confidence interval or a p-value from a paired test (e.g., McNemar’s test for paired binary outcomes) to confirm the gain is not due to random fluctuation.

These are reporting gaps that can be fixed by re-aggregating existing data or adding standard error bars, rather than requiring new experiments, provided the raw per-seed or per-trial data exists.

Action items.

- **[writing]** Table 1 (MetaWorld) and Table 2 (LIBERO) report single-point success rates (e.g., 64.35%) without any measure of uncertainty (SD, SE, or CI) or mention of the number of random seeds used. In RL/robotics, performance variance across seeds is high; reporting a single mean without spread implies false precision. Report mean $\pm$ SD over at least 3-5 seeds for all main results.

- **[writing]** Table 3 (Real-World) reports ‘ 73.3 ± 6.5 ’ as the average success rate across 9 tasks, calculating the standard deviation of the *task means* rather than the standard deviation of the *trials* or the standard error of the mean. This conflates task difficulty variance with method stability. Report the standard deviation of the 20 trials per task, or the standard error of the mean across tasks, to correctly reflect the precision of the reported average.
- **[writing]** Section 5.2 (Mechanism Analysis) claims OGG improves recovery with an ‘average gain of 0.23’ (Fig 5) but provides no statistical test (e.g., paired t-test or bootstrap) or confidence interval to support this magnitude. Given the binary nature of success/failure, report the 95% confidence interval for the difference in recovery rates or the p-value from a McNemar’s test on the paired outcomes.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and readable, with a clear logical flow from problem statement to solution. However, several sections suffer from paragraph density and minor structural inconsistencies that force the reader to re-parse sentences or infer missing transitions.

The most significant issue is in Section 3.1, where the training methodology, loss function, and justification for using demonstrations are compressed into a single, dense paragraph. This obscures the distinct components of the method. Splitting this into three focused paragraphs would significantly improve clarity. Similarly, Section 3.2 introduces the LVM components with abrupt bolded sub-headers rather than smooth transitions, making the operational flow feel disjointed.

In Section 3.4, the definition of the corrective target relies on a variable (k) that was defined in a previous section but not re-contextualized, creating a momentary ambiguity for the reader. The Introduction’s contribution list also contains a grammatical fragment in the final bullet point, which breaks the parallel structure of the list. Finally, the opening of Section 4.2 uses inconsistent punctuation and phrasing in its list of analysis perspectives, which slightly disrupts the professional tone.

Addressing these specific structural and grammatical points will allow the reader to move through the technical details without unnecessary friction.

Action items.

- **[writing]** The paper is generally well-structured and readable, with a clear logical flow from problem statement to solution. However, several sections suffer from paragraph density and minor structural inconsistencies that force the reader to re-parse sentences or infer missing transitions. The most significant issue is in Section 3.1, where the training methodology, loss function, and justification for using demonstrations are compressed into a single, dense

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